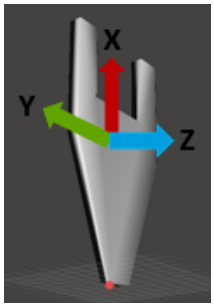
Application of Reinforcement Learning in Step Convergence for 6-DoF Robotic Arm Movement Path

Abstract.

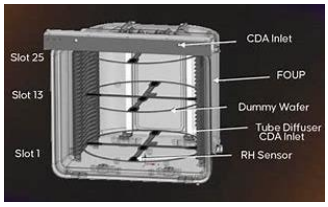
Current trajectories for industrial robotic arm are predetermined by experienced robotics engineers to ensure stability and reliability. However, those manually tuned paths may lead to reduced efficiency. In this research, we implemented reinforcement learning to guide a robotic arm to move toward target points with specific orientation by designing well-structured observation and reward functions.

Introduction

For industrial applications, robotic arm with fixed paths are preferred due to their monotonous work environment and high stability. These paths often take a lot of efforts to design and verify by experienced robotic engineers. In our task, the robotic arm need to lift FOUP up with specific positions and orientation, we implemented TD3 (Twin-Delayed DDPG) to optimize steps needed to move robotic arm to target. TD3 is often used in problems with continuous action space, which is suitable for velocity-controlled robotic arms.



- xyz orientation



- FOUP

Design

Reward function.

$$\boldsymbol{reward_{dist} = 1 - \tanh(distance)}$$

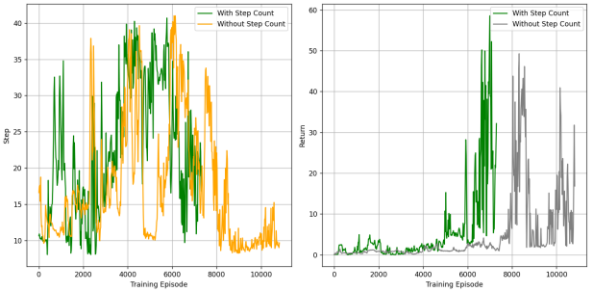
$$\boldsymbol{reward_{ori} = 1 - \tanh(\sqrt{roll_{difference}^{\,2} + pitch_{difference}^{\,2} + yaw_{difference}^{\,2}})}$$

Observation.

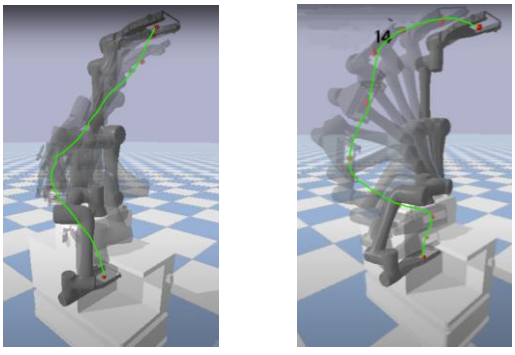
Observation is an 11-dimensional vector. Besides the robotic arm configuration, The step count of the robotic arm is also included in the observation to inform the arm of the number of steps it has taken. This aligns with the concept of **time-awareness** mentioned in the [1].

$$\begin{bmatrix} joint\ angles * 6 \\ distance_{eef\ to\ target} \\ orientation_{eef} \\ step\ count \end{bmatrix}$$

Results



a) Steps taken each episode.
(Adding step count can improve training speed)



b) Robot behavior at the end of learning, robot on the right takes less steps along the path

References

[1] J. Pardo, B. Peng, C. Kuttler, and R. Jensfelt, *Time Limits in Reinforcement Learning*, arXiv, 2017.

[2]A. Author1, B. Author2, and C. Author3, "Application of Reinforcement Learning to the Orientation and Position Control of a 6 Degrees of Freedom Robotic Manipulator," *arXiv preprint arXiv:2309.12560*, 2023.